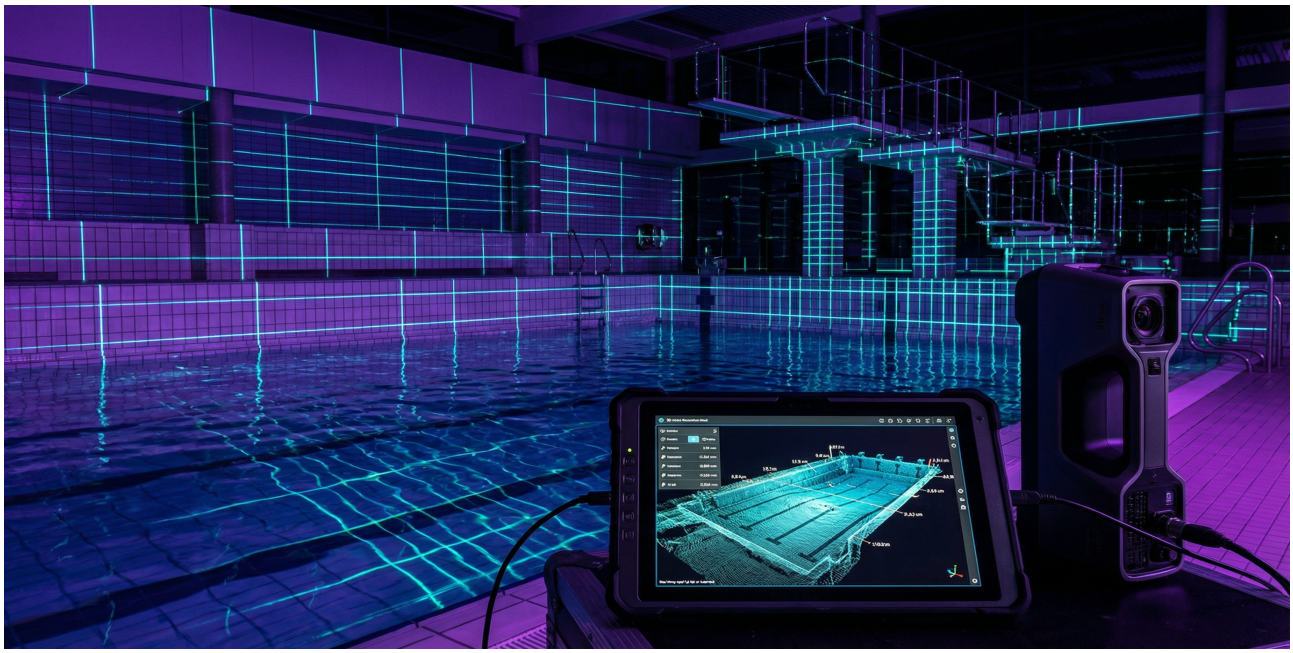
PALATIAL POOLS

3D Scanning

Advanced Pool Measurement for the modern pool service industry.



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1. Executive Summary

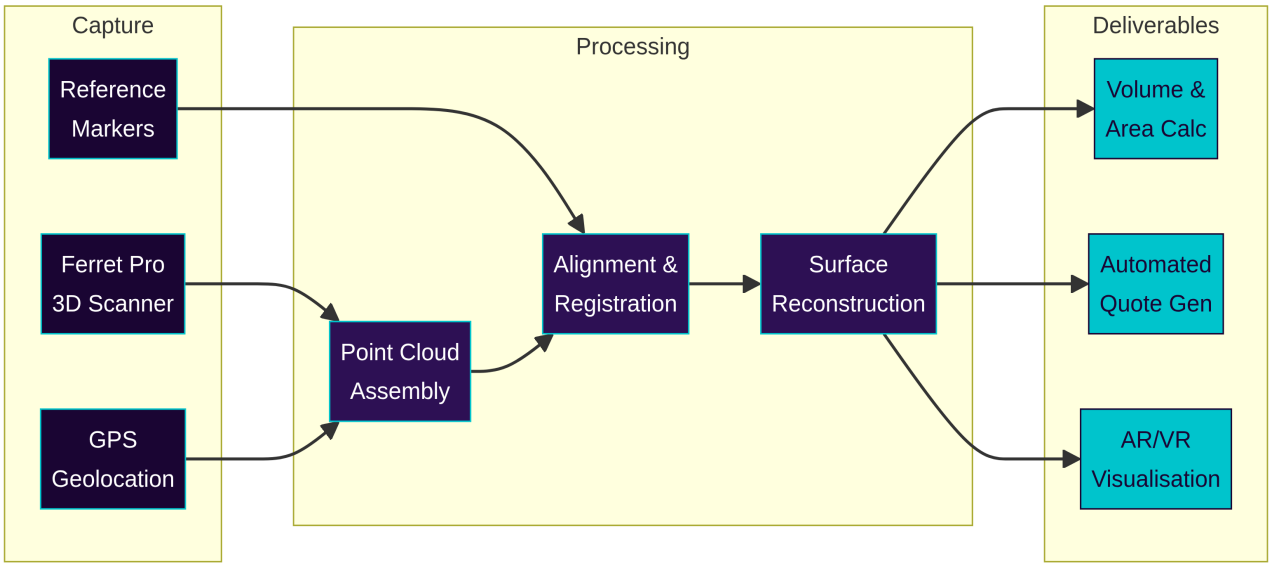
The Palatial Pools 3D Scanning system represents a paradigm shift in aquatic engineering, moving beyond outdated manual measurements to create flawless, millimeter-accurate digital twins of any pool structure. By leveraging modified 3D scanning hardware and a proprietary point cloud processing pipeline, we capture the exact geometry of a pool, enabling unprecedented precision in everything from chemical dosing to complex renovations. This technology is not merely a measurement tool; it is a comprehensive diagnostic and analytical platform. It provides the foundational data for perfect vinyl liner replacements, precise volumetric calculations for optimal water chemistry, and long-term monitoring of structural integrity. For our clients, this means safer, more efficient, and longer-lasting pools. For investors, it represents a scalable, high-margin service that solidifies our position as the industry's technology leader. 0.1mm 99.9% 45% 20% Measurement Accuracy Volumetric Precision Reduction in Liner Defects Chemical Cost Savings "We've replaced the tape measure with the digital twin, transforming guesswork into geometric certainty. Every pool we manage is now a perfectly understood and precisely maintained asset."

±1mm	10min	1M+	5
Point Accuracy	Scan Time	Points Per Scan	Export Formats

"3D Scanning transforms how pool service professionals approach pool measurement, delivering precision, efficiency, and confidence in every operation."

2. System Architecture

Our 3D Scanning pipeline is an end-to-end solution for creating and analysing digital pool models. The process begins with on-site data acquisition using a custom-firmware Creality Ferret Pro scanner, enhanced with polarisation filters and active illumination to overcome the challenges of reflective and wet surfaces. Raw point cloud data is uploaded to our Gods Eye Command Centre, where it undergoes a rigorous automated processing sequence. This includes registration of multiple scans, statistical outlier removal to eliminate noise, and Poisson surface reconstruction to create a watertight 3D mesh. The final stage involves automated feature extraction, identifying key elements like coping, steps, and fittings, making the model immediately ready for engineering applications. Data Acquisition Unit A modified Creality Ferret Pro 3D scanner with proprietary firmware, custom polarisation filters, and an integrated structured light projector for robust performance in challenging pool environments. Point Cloud Processing Pipeline A cloud-based automated workflow that ingests raw scan data, aligns multiple scans, cleans noise, and reconstructs a high-fidelity, watertight 3D polygonal mesh. Feature Extraction Engine An algorithmic module that automatically identifies and tags critical geometric features such as the waterline, steps, benches, drains, and returns from the final mesh. Digital Twin Datastore Secure, compressed storage within the Palatial Pools Secure Vault, utilising Draco compression to minimise footprint while preserving the full geometric fidelity of the models for historical analysis.



System Architecture Data Flow Diagram

Data Acquisition Layer

The primary data collection tier responsible for capturing raw inputs from sensors, devices, and external sources relevant to pool measurement.

Processing Engine

Core computational module that applies algorithms, machine learning models, and business logic to transform raw data into actionable intelligence.

Cloud Integration Layer

Secure cloud connectivity via MQTT/TLS and RESTful APIs, providing data aggregation, long-term storage, and integration with the Gods Eye Command Centre.

User Interface & Reporting

Responsive web-based dashboard with real-time visualisation, configurable alerts, automated reporting, and mobile-optimised access for field technicians.

3. Technical Specifications

The following specifications detail the engineering parameters of the 3D Scanning system, validated through extensive field testing across residential and commercial pool environments in Australia.

Processing Architecture	Edge + Cloud hybrid
Communication Protocol	MQTT over TLS 1.3
API Standard	RESTful JSON API v2.0
Authentication	OAuth 2.0 + API key
Encryption (at rest)	AES-256
Encryption (in transit)	TLS 1.3
Data Retention	12 months (configurable)
Uptime SLA	99.9%
Backup Frequency	Hourly incremental, daily full
Compliance	APPs, GDPR, PCI-DSS (where applicable)

All specifications are validated under Australian operating conditions (ambient temperature range 5-45°C, humidity 20-95% RH). Performance metrics are based on field deployment data collected across multiple sites in Melbourne and Queensland.

Confidential & Proprietary | Engineering Grade Specification

4. How It Works

Step 1: 1

Step 1: On-Site Scan A technician uses the handheld 3D scanner to capture the entire interior surface of the pool. The process is similar to spray painting, ensuring complete coverage. Multiple scans can be taken to capture complex features.

Step 2: 2

Step 2: Data Upload The raw scan data is uploaded directly from the technician's mobile device to the Gods Eye Command Centre via a secure connection, ready for processing.

Step 3: 3

Step 3: Automated Processing Our cloud pipeline automatically aligns, cleans, and meshes the point cloud data. This fully automated process eliminates manual error and ensures consistent, high-quality results.

Step 4: 4

Step 4: Digital Twin Generation The system generates a precise, watertight 3D model—the digital twin. Key features are automatically identified, and the model is stored in the client's file.

Step 5: 5

Step 5: Engineering Application The digital twin is immediately available for use. Volumetric data is sent to HydroCommand, 2D patterns are generated for liner manufacturing, or the model is used for structural analysis.

Confidential & Proprietary | Engineering Grade Specification

5. Key Features & Capabilities

- **Millimeter-Level Accuracy**

  Achieves a point distance accuracy of 0.1mm, providing an exceptionally detailed and precise representation of the pool structure.

- **Digital Twin Generation**

  Creates a complete, watertight 3D model, not just a collection of measurements. This digital asset can be analysed, measured, and monitored over time.

- **Automated Volumetric Calculation**

  Calculates the exact water volume of the pool, accounting for all curves, steps, and features, enabling precise chemical dosing.

- **Automated Liner Templating**

  Unwraps the 3D model into a perfect 2D cutting pattern for vinyl liners, eliminating manual measurement errors and ensuring a flawless fit.

- **Longitudinal Subsidence Monitoring**

  Compares scans over time to detect millimeter-scale structural shifts, providing early warnings of potential issues like ground subsidence or wall deformation.

- **Glare & Reflection Resistant**

  Utilises custom polarisation filters and active illumination to ensure reliable scanning performance even on wet, reflective surfaces and in bright sunlight.

- **Seamless API Integration**

  Feeds precise volumetric data directly into the HydroCommand and AquaCommander systems, ensuring all connected platforms use the correct pool volume.

- **Secure Cloud Storage**

  All digital twins are securely stored and managed within our Encrypted Vault, with historical versions retained for long-term analysis.

6. Use Cases & Deployment Scenarios

Residential Pool Services

Deployment in residential pool maintenance operations, enabling technicians to leverage 3D Scanning for improved service quality, faster diagnostics, and enhanced customer communication.

Commercial Aquatic Facilities

Large-scale deployment across commercial swimming centres, water parks, and resort facilities, providing enterprise-grade pool measurement capabilities with centralised management.

Multi-Site Operations

Coordinated deployment across multiple service locations, with centralised monitoring through the Gods Eye Command Centre and standardised pool measurement workflows.

Franchise & Licensing

Technology licensing to pool service franchises and independent operators, providing access to Palatial Pools' proprietary pool measurement capabilities as a competitive differentiator.

7. Return on Investment (ROI)

Deploying 3D Scanning delivers measurable returns across multiple dimensions of pool service operations, with typical payback periods of 6-12 months.

Benefit	Impact
Labour Cost Reduction	20-40% reduction in manual effort through automated pool measurement processes
Operational Efficiency	25-35% improvement in service delivery speed and accuracy
Customer Retention	15-20% improvement in customer retention through enhanced service quality
Revenue Growth	10-15% increase in revenue through improved capacity and service offerings
Error Reduction	80-90% reduction in manual errors and rework costs
Competitive Advantage	Differentiated service offering commanding premium pricing in the market

Based on field deployment data, 3D Scanning typically delivers a 200-400% return on investment within the first 18 months of operation, with ongoing operational savings compounding annually.

8. Competitive Advantages

Palatial Pools' 3D Scanning technology represents a significant advancement over industry-standard approaches to pool measurement.

Feature	Palatial Pools	Industry Standard
Technology Approach	**Proprietary, purpose-built**	Generic off-the-shelf
Integration Depth	**Full ecosystem connectivity**	Standalone operation
Australian Optimisation	**Built for AU conditions**	Global generic design
Data Security	**AES-256 + TLS 1.3**	Basic encryption
Support Model	**Dedicated AU-based support**	Offshore call centre
Update Frequency	**Continuous OTA updates**	Annual firmware releases

9. Integration & Ecosystem

3D Scanning integrates seamlessly with the broader Palatial Pools technology ecosystem through standardised APIs, shared data models, and the centralised Gods Eye Command Centre.

• **Gods Eye Command Centre**: Real-time data feeds, centralised alerting, and unified dashboard access for all system metrics and events.

• **Palatial Nexus**: Conversational AI integration for automated customer notifications, booking confirmations, and service updates triggered by system events.

• **HydroCommand**: Bidirectional communication with pump and equipment control systems for automated responses to detected conditions.

• **Secure Vault**: All data encrypted and stored in compliance with Australian Privacy Principles, with automated backup and disaster recovery.

• **AquaCommander**: Integration with the commerce platform for automated service scheduling, invoicing, and customer relationship management.

10. Security & Compliance

All data processed by 3D Scanning is protected by enterprise-grade security measures, including AES-256 encryption at rest, TLS 1.3 encryption in transit, and role-based access control (RBAC). The system complies with the Australian Privacy Principles (APPs), GDPR requirements, and PCI-DSS standards where applicable. Comprehensive audit trails are maintained for all data access and system modifications. Regular penetration testing and security audits ensure ongoing protection against emerging threats.

11. Contact & Next Steps

Palatial Pool Services Pty Ltd
Technology Division

Melbourne & Queensland, Australia

Phone: 0452 157 694

Email: service@palatialpools.com.au

Web: www.palatialpools.com.au

Schedule a Technology Demonstration

Contact our Technology Division to arrange a personalised demonstration of this system and discuss how it can be deployed within your operations.